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14 March 1974
SUPERSEDING
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MILITARY SPECIFICATION

CATHETER, URETERAL, X-RAY, WHISTLE TIP, BRAASCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ureteral catheters of the Braasch design, to be used to catheterize the ureters and for preparation of roentgenograms.

1.2 Classification. Catheters shall be of the following sizes, as specified (see 6.1):

Size 8 - 8 French.
Size 10 - 10 French.
Size 12 - 12 French.
Size 14 - 14 French.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-up, Paperboard.

Military

MIL-L-10547	Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.

3.1.1 Woven fabric. The fabric shall be woven from good quality nylon or dacron of a size appropriate for the diameter of the catheter.

3.1.2 Coating. The coating shall be a flexible varnish, or similar coating, containing x-ray opaque material.

3.2 Style, design and dimensions. Graduated x-ray ureteral catheters shall each be composed of a tightly and smoothly woven base, free from loose ends or broken fibers, and completely, uniformly, and symmetrically covered with a flexible coating containing the x-ray opaque material. The finished catheter shall have the proper flexibility, pliability, and body required to withstand the pressure of introduction through the cystoscope and into the ureter, together with the softness and pliability necessary to prevent irritation to body tissues. The area of eye and whistle tip shall be not less than the cross-sectional area of the catheter. Catheters shall be in accordance with type I or II of Figure 1.

3.3 Coating of catheter. The coating shall thoroughly and uniformly impregnate in and around the woven base in a manner to coat the braided fibers completely. The coating thickness shall be relatively uniform throughout the length of the catheter. The coating shall be formulated to secure maximum adhesion to the tube and shall have the required toughness and elasticity to permit repeated flexing and to prevent peeling, cracking, or breaking when tested in accordance with 4.4.1, 4.4.2 and 4.4.3. The coating shall be resistant to body fluids, mild acids, alkalies, and deterioration due to aging during storage. The coated catheter shall show no signs of brittleness, blistering, or other visible damage as a result of the boiling or auto-claving tests described in 4.4.1 and 4.4.2. The coated catheter shall not be tacky, shall not stick together, and its surface shall not be marred nor show any other evidence of damage as a result of the flexibility tests described in 4.4.3.

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3.4 Toxicity. The catheters shall be free from toxic material or any ingredient that might have a deleterious effect upon body tissues.

3.5 X-ray opacity. The catheter shall throw a definite, uniform shadow, when tested as described in 4.4.4. The degree of opaqueness shall be such that the shadow of a small foreign body in the front or rear of the catheter shall be visible for diagnosis.

3.6 Sizes. Catheters shall be woven on a mandrel of the applicable diameters shown in Table I. The lumen of each catheter stem shall be such that, after soaking, the catheter shall be capable of passing freely a mandrel of slightly smaller diameter, as specified in 4.4.5.

Table I. Mandrel diameters for weaving catheters

Catheter stem size (French)	Mandrel size (inch)
4	0.020
5	0.029
6	0.039
7	0.045
8	0.051

3.7 Finish. The surface of each catheter shall be properly worked down and given a dry, smooth, highly polished but elastic finish, free from scratches, grooves, or other defects which would prevent its easy passage through the cystoscope or the ureter. The surface shall be "slick" when wet and shall pass easily.

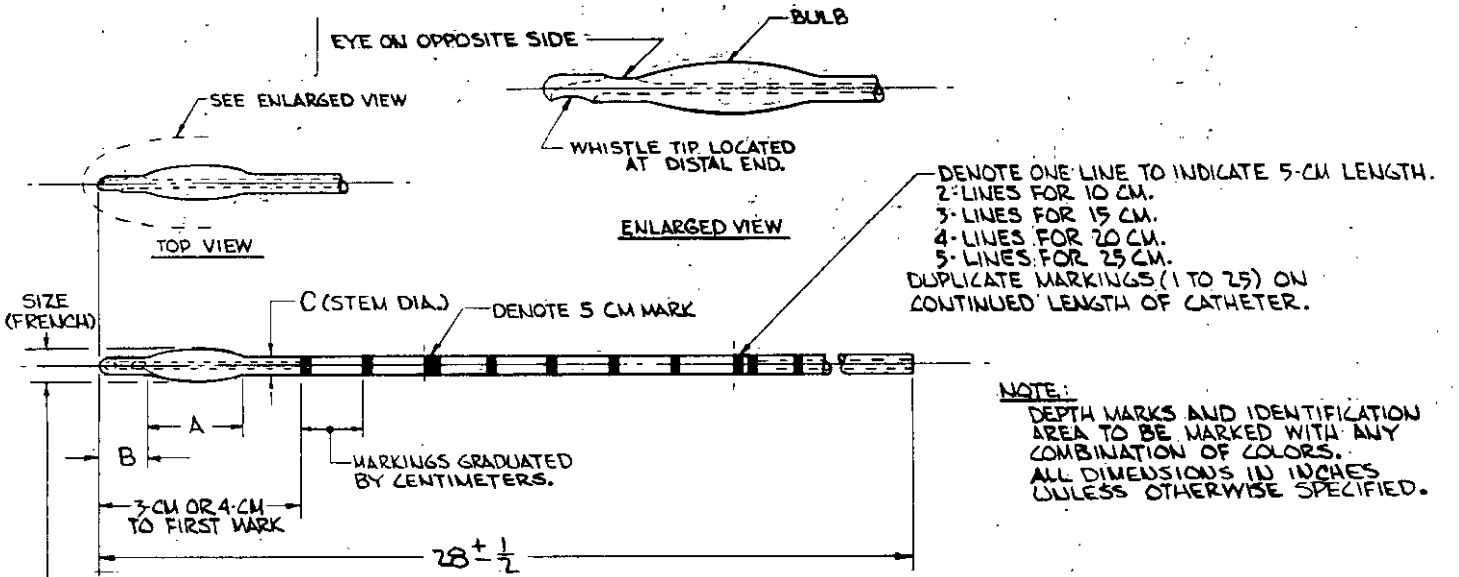
3.8 Marking. Each catheter shall be plainly and indelibly marked upon the proximal end with the manufacturer's name, its size according to the French scale, and the word "X-Ray".

3.9 Workmanship. The workmanship shall be first class throughout. The finished catheters shall be free from defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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NOTE:
DEPTH MARKS AND IDENTIFICATION AREA TO BE MARKED WITH ANY COMBINATION OF COLORS.
ALL DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

NSN	SIZE ±.45 FR.	A	B	C ± .15 FR.	
6515-00-299-8371	8 FR.	1/2	1/4	4 FR.	TYPE I
6515-00-299-8672	10 FR.	1/2	1/4	5 FR.	
6515-00-299-8679	12 FR.	1/2	5/16	5 FR.	
6515-00-299-8674	14 FR.	1/2	5/16	7 FR.	
		TOLERANCE: A+B = ± 3/32			

NSN	SIZE +0 -.45 FR.	A	B	C +0 -.3 FR.	
6515-00-299-8371	8 FR.	3/4	5/16	6 FR.	TYPE II
6515-00-299-8672	10 FR.	3/4	3/8	7 FR.	
6515-00-299-8679	12 FR.	3/4	3/8	7 FR.	
6515-00-299-8674	14 FR.	3/4	3/8	8 FR.	
		TOLERANCE: A+B = ± 1/16			

FIGURE 1

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table II. The unit of product for examination purposes shall be one catheter.

Table II. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table III. Where feasible, the same sample shall be used for the determination of two or more test chacteristics. Unit of product for test purposes shall be one catheter.

Table III. Sampling for Tests

Characteristics	Requirement	Test procedure	Inspection level	AQL (percent defective)
Boiling test	3.3	4.4.1	S-1	*
Autoclaving test	3.3	4.4.2	S-1	*
Flexibility after boiling	3.3	4.4.3.1	S-1	*
Flexibility after autoclaving	3.3	4.4.3.2	S-1	*
X-ray opacity	3.5	4.4.4	S-1	*
Size	3.6	4.4.5	S-1	*

* Acceptance number shall be zero for all sample sizes.

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4.3 Examination. The catheters shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include, but shall not be limited to the defects listed in Table IV.

Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Outside surfaces of catheter tube not smooth and free from burrs or surface imperfections.
102	Holes at distal end of catheter unit not free from burrs, slivers, or loose particles.
103	Catheter tube not free from kinks.
Minor	
201	Identification marking not clearly imprinted.
202	Dimensions not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Boiling test. Each sample catheter shall be boiled for ten 20-minute intervals, with 24 hours between each boiling to permit drying out at room temperature (68° to 72°F.).

4.4.2 Autoclaving test. Each sample catheter selected for test shall be autoclaved for ten 10-minute intervals at 250°F., with sufficient time between to permit cooling to room temperature (68° to 72°F.).

4.4.3 Flexibility tests.

4.4.3.1 After boiling. The catheter shall be twisted three full length turns around a catheter of the same size, subjected to boiling for 20 minutes, allowed to return to room temperature, then untwisted.

4.4.3.2 After autoclaving. The catheter shall be twisted three full length turns around a catheter of the same size, subjected to autoclaving for 15 minutes at 250°F., allowed to return to room temperature, then untwisted.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One catheter, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V, shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	6 units	360 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each catheter shall be packed (not coiled) in a sealed, combination polyethylene and paper, glassine or polyethylene envelope of appropriate size and design.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Each fiberboard box shall conform to the special requirements as specified in Table I of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The catheters are used for x-ray diagnosis in ureters.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required.
(c) Selection of applicable levels of packaging and packing.

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6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-299-8371	CATHETER, URETERAL, X-Rav, Whistle Tip, Braasch, 8 Fr.
6515-00-299-8672	CATHETER, URETERAL, X-Ray, Whistle Tip, Braasch, 10 Fr.
6515-00-299-8674	CATHETER, URETERAL, Whistle Tip, Braasch, 14 Fr.
6515-00-299-9679	CATHETER, URETERAL, Whistle Tip. Braasch, 12 Fr.

It does not include all types or sizes as indicated bv the title nor all types which are commercially available.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agencv - DM

Project No. 6515-1812

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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